



A Level • Cambridge (CIE) • Chemistry

56 mins 30 questions

Multiple Choice Practice

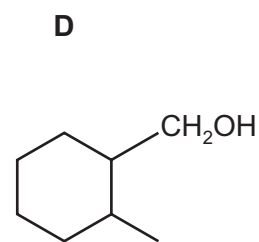
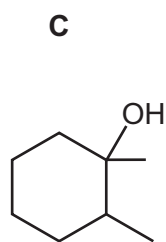
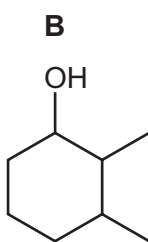
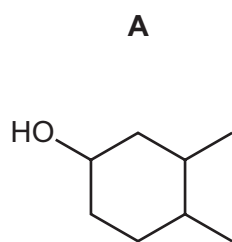
Topical Practice MCQ

16. Hydroxy compounds

Carefully selected past paper questions by topic.

- 32 Compound X is a single, pure, optical isomer. Compound X is heated with an excess of concentrated H_2SO_4 . Only **one** organic product is formed.

What is compound X?



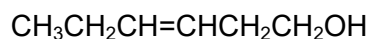
- 34** Compound Y reacts with alkaline $I_2(aq)$. When the products of this reaction are acidified, a dicarboxylic acid is produced. The formula of the dicarboxylic acid is $HOOC-R-COOH$ where R consists of one or more CH_2 groups.

Which compound is Y?

- A** pentan-1,4-diol
- B** pentan-1,5-diol
- C** pentan-2,3-diol
- D** pentan-2,4-diol

- 33 The compound 'leaf alcohol' is partly responsible for the smell of new-mown grass.

leaf alcohol



What will be formed when 'leaf alcohol' is oxidised using an excess of hot acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$?

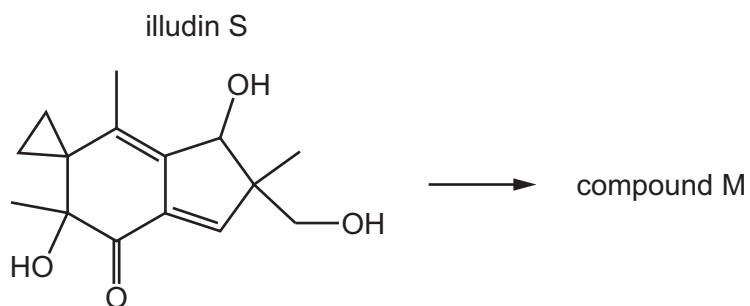
- A** $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{CO}_2\text{H}$
B $\text{CH}_3\text{CH}_2\text{COCOCH}_2\text{CO}_2\text{H}$
C $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CO}_2\text{H}$
D $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$ and $\text{HO}_2\text{CCH}_2\text{CO}_2\text{H}$

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- A** pentan-1,4-diol
- B** pentan-1,5-diol
- C** pentan-2,3-diol
- D** pentan-2,4-diol

- 28 When illudin S is heated under reflux with an excess of acidified potassium dichromate(VI), compound M is formed.



What is the molecular formula of compound M?

- A** $\text{C}_{15}\text{H}_{16}\text{O}_4$ **B** $\text{C}_{15}\text{H}_{16}\text{O}_5$ **C** $\text{C}_{15}\text{H}_{18}\text{O}_4$ **D** $\text{C}_{15}\text{H}_{18}\text{O}_5$

34 When ethanol reacts with sodium metal, ethoxide ions, $\text{CH}_3\text{CH}_2\text{O}^-$, are produced.

When water reacts with sodium metal, OH^- ions are produced.

Which statement about these reactions and the ethoxide ion is correct?

- A** At the same temperature, the rate of reaction between sodium and ethanol is greater than that between sodium and water.
- B** $\text{CH}_3\text{CH}_2\text{O}^-$ is a stronger base than OH^- due to the electron-donating effect of the ethyl group.
- C** The negative charge on the oxygen in an ethoxide ion is delocalised.
- D** It is easier to deprotonate ethanol as it is more acidic than water.

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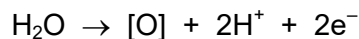
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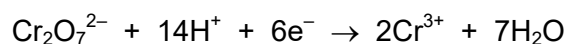
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- C** The negative charge on the oxygen in an ethoxide ion is delocalised.
- D** It is easier to deprotonate ethanol as it is more acidic than water.

- 25** When an organic compound is oxidised, any oxygen atom gained by the organic molecule is considered to be from a water molecule also producing $2\text{H}^+ + 2\text{e}^-$. Any hydrogen atom lost may be considered to be lost as $\text{H}^+ + \text{e}^-$.

These changes can be represented by the following two equations.



Compound X is oxidised by heating under reflux with hot, acidified potassium dichromate(VI) for one hour. The half-equation for the reduction reaction is shown.



Under these conditions, one mole of potassium dichromate(VI) oxidises three moles of X.

What could X be?

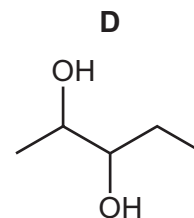
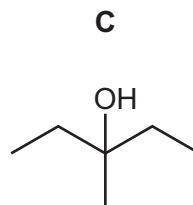
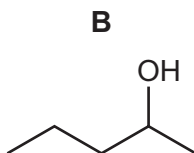
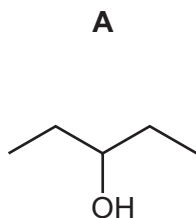
- A** propanal
- B** propan-1-ol
- C** propan-1,2-diol
- D** propan-1,3-diol

27 Structural and stereoisomerism should be taken into account when answering this question.

An organic compound, X, is dehydrated by heating with concentrated phosphoric(V) acid.

Only **two** organic products are formed.

What could be X?

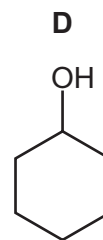
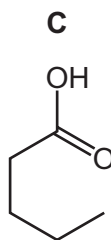
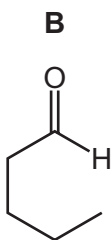
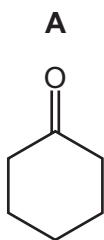


28 When compound X is heated with $\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$, a colour change from orange to green is observed.

Two tests are carried out on the organic product of this reaction.

test	result
Tollens' reagent	no change
2,4-dinitrophenylhydrazine	orange precipitate

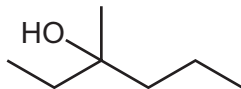
What could be compound X?



26 Structural isomerism and stereoisomerism should be considered when answering this question.

3-methylhexan-3-ol reacts with hot, concentrated sulfuric acid to form several isomeric compounds with the molecular formula C_7H_{14} .

3-methylhexan-3-ol



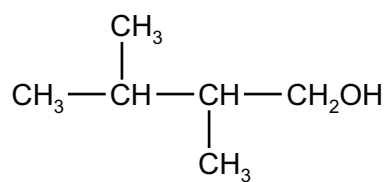
How many isomeric compounds could be formed in this reaction?

- A** 3 **B** 4 **C** 5 **D** 6

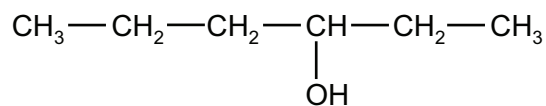
26 Structural isomerism and stereoisomerism should be considered when answering this question.

Which isomer of $\text{C}_6\text{H}_{13}\text{OH}$ gives the greatest number of different alkenes when it is dehydrated?

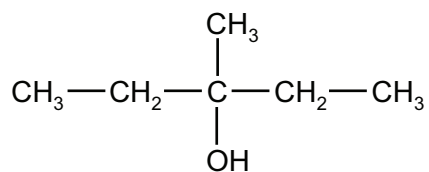
A



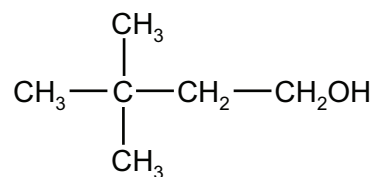
B



C



D



24 Structural isomerism and stereoisomerism should be considered when answering this question.

The molecular formula of compound X is $C_5H_{12}O$.

Compound X:

- reacts with alkaline aqueous iodine
- can be dehydrated to form two alkenes only.

What could be the identity of compound X?

- A** $CH_3CH_2CH(CH_3)CH_2OH$
- B** $CH_3CH_2CH(OH)CH_2CH_3$
- C** $(CH_3)_2CHCH(OH)CH_3$
- D** $CH_3CH_2CH_2CH(OH)CH_3$

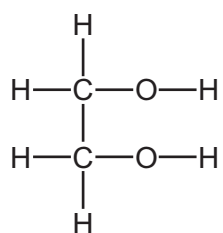
- 26** When 0.0075 mol of alcohol X are completely burnt in excess oxygen and the gases produced are passed through an excess of limewater (calcium hydroxide solution), 3.0 g of calcium carbonate are produced.

When X is warmed with acidified potassium dichromate(VI) there is a colour change from orange to green.

What could be the identity of X?

- A** $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$
- B** $(\text{CH}_3)_3\text{COH}$
- C** $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- D** $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$

27 Ethane-1,2-diol has the following structure.



Without breaking the C–C bond, there are five possible oxidation products.

What is the **total** number of aldehyde groups and carboxylic acid groups in these five products?

	–CHO	–COOH
A	3	3
B	3	4
C	4	3
D	4	4

- 27** An unknown organic compound reacts with sodium to give a combustible gas as one product but does **not** give a yellow precipitate with alkaline aqueous iodine.

What is a possible identity of the unknown organic compound?

- A** propanal
- B** propan-1-ol
- C** propan-2-ol
- D** propanone

24 Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$, causes pain when it builds up in muscles.

Which reagent reacts with **both** of the $-\text{OH}$ groups in lactic acid?

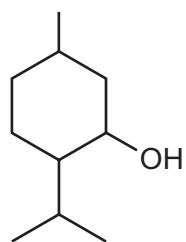
- A acidified potassium dichromate(VI)
- B ethanol
- C sodium
- D sodium hydroxide

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- A acidified potassium dichromate(VI)
- B ethanol
- C sodium
- D sodium hydroxide

25 The compound shown is menthol, a naturally-occurring alcohol found in peppermint oil.

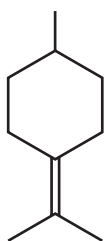


menthol

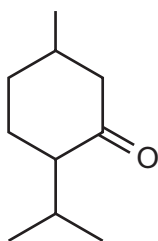
When menthol is heated with concentrated sulfuric acid it reacts. The products that form include compound **T**.

What could be the structure of compound **T**?

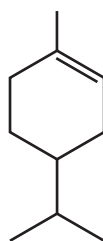
A



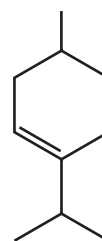
B



C

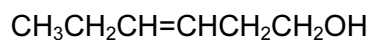


D



- 26 Which compound gives an organic product with a lower boiling point when it is heated under reflux with an excess of acidified potassium dichromate(VI)?
- A 2-methylbutan-1-ol
 - B 2-methylbutan-2-ol
 - C pentan-1-ol
 - D pentan-2-ol

- 22 The compound 'leaf alcohol' is partly responsible for the smell of new-mown grass.

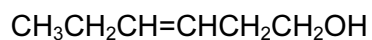


leaf alcohol

What will be formed when 'leaf alcohol' is oxidised using an excess of hot, acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$?

- A $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{CO}_2\text{H}$
- B $\text{CH}_3\text{CH}_2\text{COCOCH}_2\text{CO}_2\text{H}$
- C $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CO}_2\text{H}$
- D $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$ and $\text{HO}_2\text{CCH}_2\text{CO}_2\text{H}$

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- B $\text{CH}_3\text{CH}_2\text{COCOCH}_2\text{CO}_2\text{H}$
- C $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CO}_2\text{H}$
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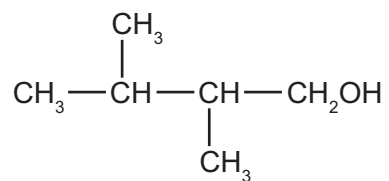
- 24** Complete combustion of compound **X** produces carbon dioxide and water only. **X** produces steamy fumes with PCl_5 . **X** does not give any visible product with 2,4-dinitrophenylhydrazine reagent.

What can be deduced with certainty from this information?

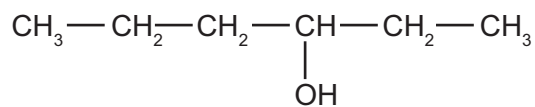
- A** **X** is a carboxylic acid.
- B** **X** is a hydrocarbon.
- C** **X** is an alcohol.
- D** **X** is not an aldehyde.

26 Which isomer of $\text{C}_6\text{H}_{13}\text{OH}$ gives the greatest number of different alkenes when it is dehydrated?

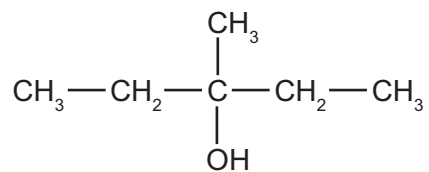
A



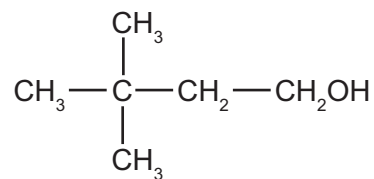
B



C



D



25 *Use of the Data Booklet is relevant to this question.*

2.76 g of ethanol were mixed with an excess of aqueous acidified potassium dichromate(VI). The reaction mixture was then boiled under reflux for one hour. The organic product was then collected by distillation.

The yield of product was 75.0 %.

What mass of product was collected?

- A** 1.98 g **B** 2.07 g **C** 2.70 g **D** 4.80 g

26 *Use of the Data Booklet is relevant to this question.*

2.30 g of ethanol were mixed with aqueous acidified potassium dichromate(VI). The desired product was collected by immediate distillation under gentle warming.

The yield of product was 70.0%.

What mass of product was collected?

- A** 1.54 g **B** 1.61 g **C** 2.10 g **D** 3.14 g

29 Which isomer of $\text{C}_4\text{H}_{10}\text{O}$ forms three alkenes on dehydration?

- A** butan-1-ol
- B** butan-2-ol
- C** 2-methylpropan-1-ol
- D** 2-methylpropan-2-ol

27 Which isomer of $C_4H_{10}O$ forms three alkenes on dehydration?

- A** butan-1-ol
- B** butan-2-ol
- C** 2-methylpropan-1-ol
- D** 2-methylpropan-2-ol

- 21 What will react differently with the two isomeric alcohols, $(\text{CH}_3)_3\text{CCH}_2\text{OH}$ and $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{OH}$?
- A acidified aqueous potassium manganate(VII)
 - B concentrated sulfuric acid
 - C phosphorus pentachloride
 - D sodium

30 Which isomer of $\text{C}_4\text{H}_{10}\text{O}$ forms three alkenes on dehydration?

- A butan-1-ol
- B butan-2-ol
- C 2-methylpropan-1-ol
- D 2-methylpropan-2-ol

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response

Answer Key

No.	Paper	Topic	Answer
1	9701_s24_qp_11	16. Hydroxy compounds	D
2	9701_w24_qp_11	16. Hydroxy compounds	D
3	9701_w24_qp_12	16. Hydroxy compounds	C
4	9701_w24_qp_13	16. Hydroxy compounds	D
5	9701_m23_qp_12	16. Hydroxy compounds	B
6	9701_w23_qp_11	16. Hydroxy compounds	B
7	9701_w23_qp_13	16. Hydroxy compounds	B
8	9701_s21_qp_11	16. Hydroxy compounds	A
9	9701_w21_qp_12	16. Hydroxy compounds	A
10	9701_m19_qp_12	16. Hydroxy compounds	D
11	9701_s19_qp_11	16. Hydroxy compounds	C
12	9701_s19_qp_13	16. Hydroxy compounds	B
13	9701_m18_qp_12	16. Hydroxy compounds	C
14	9701_s18_qp_13	16. Hydroxy compounds	A
15	9701_m17_qp_12	16. Hydroxy compounds	D
16	9701_s16_qp_13	16. Hydroxy compounds	B
17	9701_w14_qp_11	16. Hydroxy compounds	C
18	9701_w14_qp_12	16. Hydroxy compounds	C
19	9701_s13_qp_12	16. Hydroxy compounds	D
20	9701_s13_qp_13	16. Hydroxy compounds	D
21	9701_w13_qp_11	16. Hydroxy compounds	C
22	9701_w13_qp_12	16. Hydroxy compounds	C
23	9701_w12_qp_13	16. Hydroxy compounds	D
24	9701_s11_qp_13	16. Hydroxy compounds	B
25	9701_w11_qp_11	16. Hydroxy compounds	C
26	9701_w11_qp_12	16. Hydroxy compounds	A
27	9701_s10_qp_11	16. Hydroxy compounds	B
28	9701_s10_qp_12	16. Hydroxy compounds	B

Answer Key

No.	Paper	Topic	Answer
29	9701_s10_qp_13	16. Hydroxy compounds	B
30	9701_s10_qp_13	16. Hydroxy compounds	B